Supplementary Evidence for Explainable Early Warning for Next-Week Abnormal Reported 311 Demand

This supplement is generated from the repository's current stored artifacts. The final comparison uses the T2 event: a shifted eight-week abnormality and next-week count at least three. Unless stated otherwise, the final protocol trains through 2023, selects calibration and thresholds on 2024, and evaluates unchanged on held-out 2025 rows.

S1. Historical-volume performance

Train-derived cut points of the shifted eight-week historical mean define the volume groups; no test label is used to form the groups. Repeated zero-valued cut points collapse the nominal deciles to nine observable bins. All rows below sum to the full held-out 2025 T2 population.

Volume group	Rows	Prevalence	PR-AUC	Precision	Recall	F1	P@5%	Alert rate
D1	22,986	1.0%	0.085	0.142	0.070	0.094	0.071	0.5%
D2	11,784	11.1%	0.249	0.227	0.538	0.319	0.341	26.2%
D3	10,915	14.1%	0.295	0.238	0.639	0.346	0.410	37.9%
D4	11,920	14.8%	0.325	0.274	0.584	0.373	0.455	31.5%
D5	12,453	14.4%	0.311	0.266	0.568	0.363	0.409	30.7%
D6	12,491	12.6%	0.277	0.247	0.556	0.342	0.368	28.3%
D7	11,129	12.7%	0.310	0.264	0.547	0.356	0.427	26.3%
D8	11,432	13.3%	0.339	0.277	0.618	0.382	0.458	29.6%
D9	17,506	13.9%	0.399	0.304	0.599	0.403	0.553	27.5%

Table 1: Final Platt-calibrated LightGBM performance by train-derived historical-volume group on held-out 2025 T2 rows.

S2. Complaint-category mapping and residual category

Analysis category	Types	Requests	Share
housing	57	8,061,962	26.06%
noise	9	6,713,922	21.70%
parking/traffic	39	6,521,679	21.08%
sanitation	36	3,090,901	9.99%
infrastructure	28	2,673,125	8.64%
public/safety	38	1,597,023	5.16%
water/sewer	10	1,212,939	3.92%
environment	17	626,930	2.03%
other	86	441,648	1.43%

Table 2: Deterministic mapping of original 311 complaint types to the nine analysis categories.

Complaint type	Requests	Share of other
Encampment	191,408	43.34%
Drug Activity	93,873	21.26%
Animal in a Park	29,914	6.77%
Vending	27,575	6.24%
Electronics Waste	26,487	6.00%
Miscellaneous Categories	13,752	3.11%
Unleashed Dog	11,718	2.65%
DOF Property - Payment Issue	11,414	2.58%
DRIE	4,023	0.91%
Storm	3,003	0.68%
Posting Advertisement	2,745	0.62%
DPR Internal	2,520	0.57%

Table 3: Largest original complaint types in the residual `other` category.

S3. Same-target baseline comparison

Every row uses the same held-out 2025 T2 population. Formula-threshold decisions apply the T2 threshold to predicted counts; the LightGBM threshold is selected on validation 2024.

Method	Decision	Count MAE	PR-AUC	P@5%	Precision	Recall	F1
Poisson GLM, no NTA FE	Formula threshold	9.069	0.141	0.146	0.376	0.038	0.068
Poisson GLM + NTA FE	Formula threshold	9.070	0.141	0.146	0.375	0.038	0.068
HGB Poisson count	Formula threshold	8.021	0.153	0.176	0.513	0.103	0.171
Hurdle HGB count	Formula threshold	7.938	0.152	0.172	0.497	0.111	0.181
No-shortcut LightGBM classifier	Validation threshold	–	0.317	0.418	0.263	0.574	0.361

Table 4: Full baseline metrics on the shared T2 task.

S4. Bootstrap uncertainty and seed stability

Baseline intervals use 250 NTA-category cluster resamples; the main LightGBM confidence intervals reported in the manuscript use 1,000. The reported estimate is the held-out 2025 value; this table does not turn distinct decision rules into a single superiority claim.

Model	Metric	Estimate	Cluster-bootstrap 95% CI
No-shortcut LightGBM	PR-AUC	0.317	[0.308, 0.326]
No-shortcut LightGBM	P@5%	0.418	[0.403, 0.432]
No-shortcut LightGBM	F1	0.361	[0.355, 0.367]
Poisson GLM, no NTA FE	PR-AUC	0.141	[0.138, 0.144]
Poisson GLM, no NTA FE	P@5%	0.146	[0.136, 0.155]
Poisson GLM, no NTA FE	F1	0.068	[0.062, 0.076]
Poisson GLM + NTA FE	PR-AUC	0.141	[0.138, 0.144]
Poisson GLM + NTA FE	P@5%	0.146	[0.135, 0.157]
Poisson GLM + NTA FE	F1	0.068	[0.061, 0.075]
HGB Poisson count	PR-AUC	0.153	[0.149, 0.157]
HGB Poisson count	P@5%	0.176	[0.165, 0.184]
HGB Poisson count	F1	0.171	[0.162, 0.180]
Hurdle HGB count	PR-AUC	0.152	[0.148, 0.156]
Hurdle HGB count	P@5%	0.172	[0.164, 0.183]
Hurdle HGB count	F1	0.181	[0.171, 0.189]

Table 5: Cluster-bootstrap uncertainty for final-comparison rows.

Configuration	Seeds	PR-AUC	P@5%	F1
03 history calendar	5	0.288 ± 0.003	0.385 ± 0.004	0.346 ± 0.003
04 history calendar weather	5	0.298 ± 0.003	0.390 ± 0.004	0.353 ± 0.002
07 final no shortcut borough	5	0.319 ± 0.002	0.419 ± 0.003	0.364 ± 0.002
08 final no shortcut nta fe	5	0.319 ± 0.001	0.421 ± 0.001	0.364 ± 0.001

Table 6: Five-seed held-out stability for key ablation configurations. Values are mean $\pm$ standard deviation.

S5. Chronological, calibration, and target-sensitivity evidence

Test year	Rows	Prevalence	PR-AUC	P@5%	F1
2021	122,616	11.4%	0.312	0.427	0.375
2022	122,616	10.9%	0.281	0.373	0.349
2023	122,616	10.9%	0.281	0.381	0.340
2024	124,974	11.0%	0.308	0.408	0.366
2025	122,616	11.1%	0.317	0.418	0.361

Table 7: Expanding-window rolling-origin test metrics for T2.

Method	Brier	ECE	Log loss	PR-AUC
Uncalibrated	0.168	0.249	0.493	0.317
Platt	0.087	0.006	0.293	0.317
Isotonic	0.087	0.004	0.292	0.307

Table 8: Final-style 2025 calibration diagnostics for T2.

The following is a labelled 2024–2025 target-definition diagnostic, not the same-task baseline comparison in Section S3.

Target	Rows	Prevalence	PR-AUC	P@5%	F1
T0	247,590	12.8%	0.310	0.424	0.360
T1	247,590	11.7%	0.313	0.418	0.364
T2	247,590	11.0%	0.316	0.415	0.366
T3	189,423	13.7%	0.324	0.441	0.371

Table 9: Target-sensitivity diagnostic with prevalence and ranking metrics.

S6. Count-model limitation

Poisson GLM fits converged within the configured 500 iterations. A Negative Binomial GLM is not included because `statsmodels` was unavailable in the recorded environment and scikit-learn does not provide a Negative Binomial GLM. This is a documented tooling limitation, not an omitted result.